REPTRAN and KDIS: simulating a spectral band with a few wavelengths

Two band parameterizations that SMART-G offers for molecular (gas) absorption. A reference for the SatelliteArray project: the idea behind them, what their data files contain, how SMART-G uses them, how they differ, and how this project uses them.

1. The problem

A sensor channel (a camera band) measures light averaged over a range of wavelengths. Across that range, absorption by gases (O_2, H_2O, O_3, CO_2 …) varies enormously from one wavelength to the next, over thousands of narrow lines. A direct ("line-by-line") calculation would simulate thousands of wavelengths per channel. REPTRAN and KDIS replace this by **a handful of simulations per channel**, combined with weights, while still giving the correct channel-integrated absorption.

2. The shared idea: the correlated-k method

For a band of width $\Delta\nu$ and an amount of gas u along a path, the transmission averaged over the band is

$$T = (1/\Delta\nu) \int \exp(-k(\nu)\cdot u)\, d\nu = \int_0^1 \exp(-k(g)\cdot u)\, dg \approx \sum_j w_j \cdot \exp(-k_j \cdot u)$$

The first integral runs over wavelength, where k jumps up and down (Figure 1, left). The average does not depend on *where* in the band each value of k occurs, only on how often it occurs. Sorting the values from weakest to strongest gives the smooth curve k(g), where g is the fraction of the band with absorption at most k (Figure 1, right). A few points g_i with weights w_i then integrate it accurately.

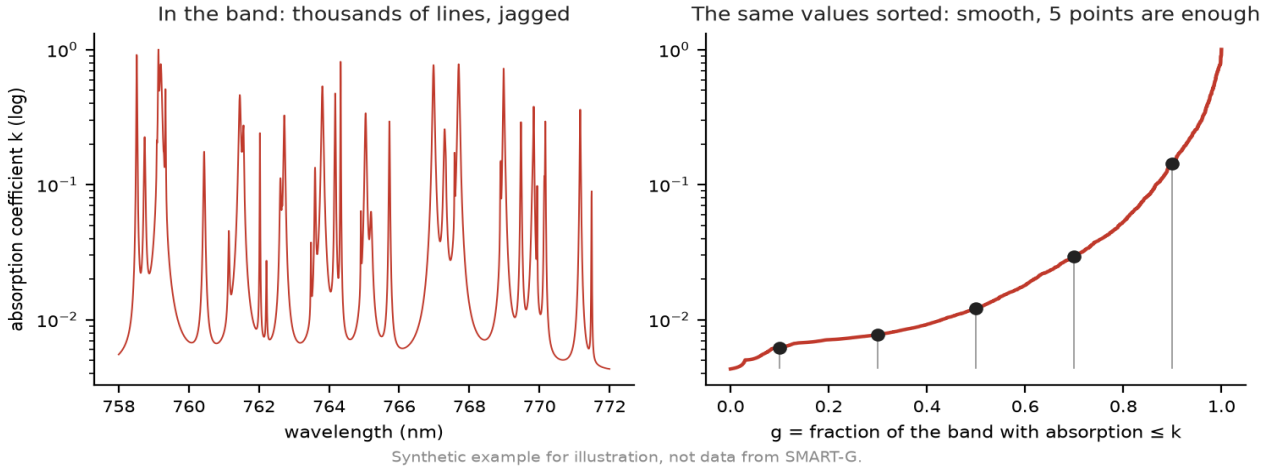


Figure 1. The sorting trick (synthetic example).

Why "correlated". In an atmosphere with layers, k also depends on pressure and temperature. The method assumes that the *order* of the wavelengths (which are strongly and which weakly absorbing) is the same at every height. Then the same point g_j refers to the same rank in all layers, so one radiative-transfer calculation per g_j, with its absorption profile through the whole atmosphere (clouds and scattering included), can be combined with the weights. It works less well where lines of different gases overlap or where scattering changes strongly across the band.

3. REPTRAN

REPTRAN (*representative wavelengths* absorption parameterization; Buehler et al. 2010, Gasteiger et al. 2014) comes from libRadtran, where it is the default spectral approach. For each band or sensor channel it picks a few **actual wavelengths**, the representative wavelengths, whose absorption stands in for the whole band, with a weight each. According to the libRadtran manual, solar bands are parameterized from 395 to 5000 nm (with single-wavelength 15 cm^{-1} bands added from 240 to 395 nm) and thermal bands from 2.5 to 100 µm.

What a REPTRAN file contains

Item (in the file)	Content
`reptran_solar_<set>.cdf`	For every band or channel: the internal wavelength grid (`wvl`, nm), the indices of its representative wavelengths (`iwvl`), their weights (`iwvl_weight`), the number of internal bands per channel (`nwvl_in_band`), the channel's width (`wvl_integral`), the extraterrestrial solar irradiance at each wavelength (`extra`, mW/(m^2 nm)), and flags saying which gases absorb there (`cross_section_source`).
`reptran_solar_<set>.lookup.<gas>.cdf`	One table per gas (H_2O, CO_2, O_3, N_2O, CO, CH_4, O_2, N_2): absorption cross-sections (in units of 10^{-20} m²) over pressure and temperature departure from a reference profile and, for H_2O, over the water-vapour mixing ratio.

The sets in the SMART-G auxiliary data

Set	Bands / channels	Representative wavelengths	Notes
coarse	5 423	7 563	generic, 15 cm^{-1} wide bands (the libRadtran default); at most 22 internal bands per band
medium	8 533	14 277	generic, 5 cm^{-1}
fine	27 193	48 939	generic, 1 cm^{-1}
sentinel	40	88	sensor channels (Sentinel-2 MSI, Sentinel-3 OLCI, …)
modis, landsat, msg, himawari, goes-r/s/t/u, parasol, proba, spot, adeos, alos, earthcare, envisat, ers, mtg-i, seosat			sensor channels, one file each; plus *thermal* sets for the infrared

How SMART-G uses it

- Choose the channels: `Reptran('reptran_solar_modis').to_smartg(band_indices=[...])` (or `include=...`, `lmin`, `lmax`) returns the internal bands, sorted by wavelength.
- Pass them as the `wavelength` of `Smartg.run` (wrapped in a `BandSet` for `atmosphere.calc`). For each internal band SMART-G computes the gas absorption profile from your atmosphere (`calc_profile`: the lookup tables interpolated in pressure and temperature, times the gas densities) and evaluates the scattering at the band's representative wavelength.
- The Monte Carlo then gives one result per internal band, and `reduce_reptran` combines them into channel values with the weights.

In `reduce_reptran` each internal band gets the factor $w \cdot \Delta\lambda$ (or $w \cdot E_0 \cdot \Delta\lambda$ with `use_solar=True`, E_0 the solar irradiance there), and the sum over a channel is divided by $\sum w \cdot \Delta\lambda$ (or by $\sum w$ with `integrated=True`). An optional `extern_weights` array adds further wavelength-dependent weights.

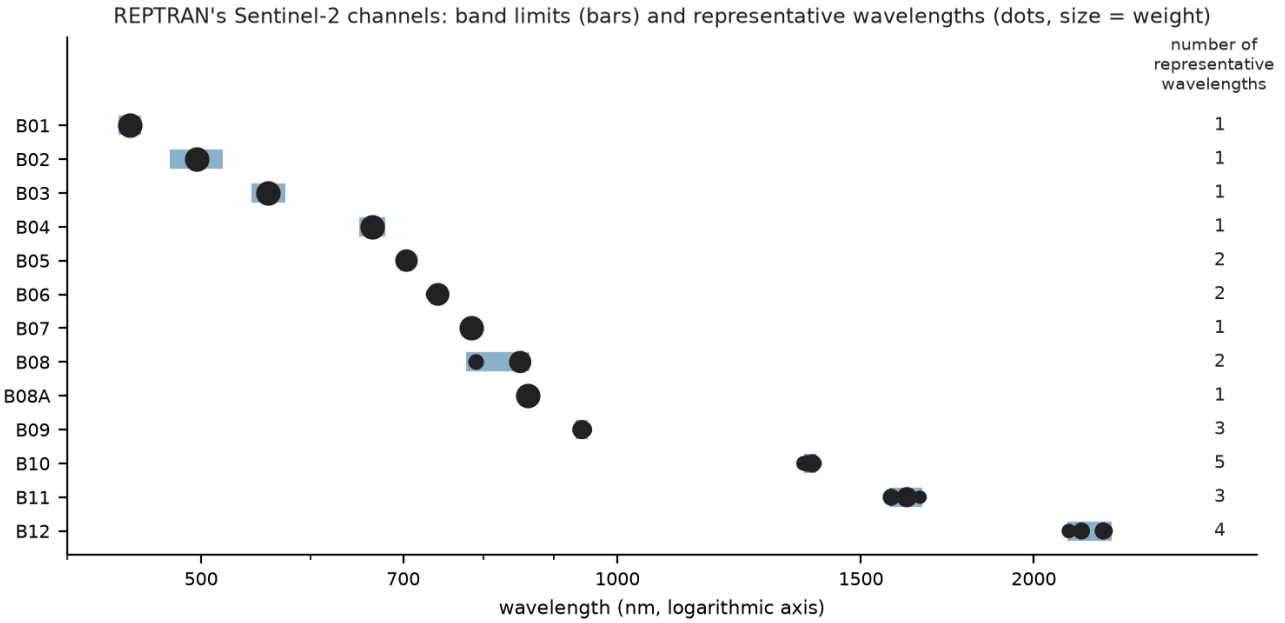


Figure 2. Real data: the REPTRAN Sentinel-2 channels. The visible bands (B01–B04) need a single representative wavelength; the bands near absorption features need more (B09 water vapour: 3, B10: 5).

4. KDIS

KDIS is the correlated-k formulation itself (SMART-G's code refers to the ARTDECO variables, and its solar-flux files are marked as produced at HYGEOS). Its definition files tabulate **k-coefficients** over pressure, temperature and, for selected gases, concentration. Each internal band is a quadrature point g_j of the k-distribution with a weight, not a distinct wavelength.

Model (in `kdis/`)	Channels	Range (nm)	Gases	Internal bands per channel
kato, kato2	32	240 – 4606	CO_2, O_2, O_3; H_2O depends on concentration	1 – 50 (median 5)
SENTINEL2_1_MSI	13	433 – 2286	CH_4, CO, CO_2, N_2, N_2O, NO_2, O_2, O_3, SO_2; H_2O by concentration	1 – 100 (median 5)
SENTINEL3_1_SLSTR	9	545 – 12 475	as Sentinel-2	1 – 2500 (median 20)

The tables have 31 pressure levels (0.007 to 1099), 9 temperatures (160 to 320 K) and 12 concentration values. Each channel stores one solar irradiance and one bandwidth. The `kato` model splits the solar spectrum into 32 bands, following Kato et al. (1999) according to the libRadtran manual, with solar fluxes, per the file name apparently from Kurucz (1992), in the accompanying `solrad_kdis_*_kurudz92.dat` files. In SMART-G the classes mirror REPTRAN's (`Kdis`, `KdisBand`, `KdisIband`, `KdisIbandList`, `reduce_kdis`), and KDIS channels have no names, so they are picked by index or by wavelength limits.

A visible difference from REPTRAN. All internal bands of a KDIS channel share the channel-centre wavelength: for Sentinel-2 B09 the five internal bands are all at 945 nm and differ only in their absorption profile, whereas REPTRAN uses three real wavelengths (940.1, 942.3 and 945.5 nm). The scattering is therefore evaluated at one wavelength per channel in KDIS.

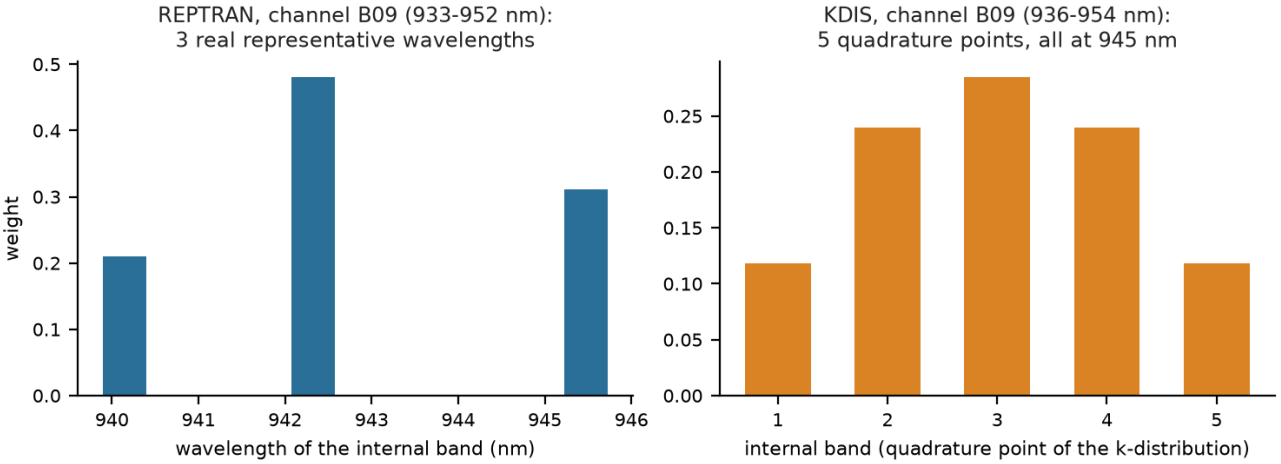


Figure 3. The same Sentinel-2 channel, B09, in the two parameterizations (real data from the SMART-G files).

5. REPTRAN and KDIS side by side

	REPTRAN	KDIS
Origin	libRadtran (Buehler et al. 2010; Gasteiger et al. 2014).	The correlated-k / k-distribution method; data files marked as produced at HYGEOS.
Internal band	A real representative wavelength with a weight.	A quadrature point of the k-distribution with a weight; shares the channel-centre wavelength.
Coverage	About 20 sensor groups, three generic resolutions, solar and thermal.	kato / kato2 (32 solar bands), Sentinel-2 MSI, Sentinel-3 SLSTR.
Absorption data	One lookup table per gas and band set (pressure, temperature, H_2O mixing ratio).	k-coefficients in one file (pressure, temperature, concentration).
Channel selection	By name (`include`), wavelength limits or index.	By index or wavelength limits (no names).
Use in SMART-G	`Reptran.to_smartg`, `reduce_reptran`, thermal: `reptran_emission`	`Kdis.to_smartg`, `reduce_kdis`, thermal: `kdis_emission`

6. What they need

- **The band definition:** a REPTRAN `.cdf` file, or a KDIS `.h5` file (with the solar flux data).
- **The absorption tables:** the REPTRAN `.lookup.<gas>.cdf` files, or the tables inside the KDIS file.
- **An atmospheric profile** per altitude layer with pressure, temperature, air density, and the densities of H_2O, CO_2, O_3, NO_2, CO, CH_4, O_2 and N_2. The AFGL profile this project uses (`atmospheres/afglus.nc`) has all of them.
- Nothing extra for the scattering: molecules, clouds and the surface are evaluated at the representative wavelength with the usual inputs.

7. In this project

The playground's `band_set` option uses REPTRAN. With `"modis"` the three channels are the MODIS Terra channels b03 (blue, 456–474 nm, one representative wavelength at 464.8 nm), b04 (green, 545–565 nm, 555.3 nm) and b01 (red, 625–667 nm, 645.9 nm). They are simulated as REPTRAN bands and combined with their weights times the solar irradiance. Because each has a single representative wavelength, a render costs the same as three plain wavelengths, and in tests the result matched monochromatic runs at those wavelengths within the Monte Carlo noise (mean reflectance 0.135 / 0.102 / 0.093 for both). For the visible range, REPTRAN's treatment of the gases therefore changes little; its value grows for bands near absorption lines. KDIS is not used by the project.

8. Limits and open points

- The channel definitions and weights are fixed in the files. There is no built-in way to give your own filter curve; for a custom band, the usual approach is to choose wavelengths and weights yourself.
- I believe, but did not verify in the data, that a sensor channel's weights include that sensor's spectral response function.
- The correlated-k assumption (same ranking of wavelengths at all heights, and smooth scattering across the band) is an approximation; it degrades where gas lines overlap or scattering varies strongly within a band.
- These parameterizations only reduce the *number of wavelengths*. They do not change how photons are sampled (see the separate note on ALIS).

References and sources

• Buehler et al. (2010), as cited by the libRadtran manual (the full reference was not available here).
• J. Gasteiger, C. Emde, B. Mayer, R. Buras, S. A. Buehler and O. Lemke, *Representative wavelengths absorption parameterization applied to satellite channels and spectral bands*, J. Quant. Spectrosc. Radiat. Transfer 148, 99–115 (2014), doi:10.1016/j.jqsrt.2014.06.024 (as cited in SMART-G's reptran module).
• libRadtran user’s guide, chapter on molecular absorption (REPTRAN resolutions, spectral ranges, Kato et al. 1999 bands).
• SMART-G 2.0.0b1: `reptran.py`, `kdis.py` and the files in `smartg_auxdata/reptran` and `smartg_auxdata/kdis`; the numbers in this note were read from those files. The citations were not checked against the originals.